

FATE P03, P04

Algorithmic Fairness I

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Fairness, Accountability, Transparency
and Ethics of Data Processing (FATE)

Programming Sessions



1. **6 Labs** = 3 Modules x 2 Sessions

2. **Deliverables per Module**

A zip file containing the notebook (worth 10 points) submitted through Aula Global

3. **Grading Policy**

~~M1: Data Anonymization (35%) Submit by 11:59 PM, Feb 02, 2026~~

M2: Algorithmic Fairness I (30%) Submit by 11:59 PM, Feb 19, 2026

M3: Algorithmic Fairness II (35%) Submit by 11:59 PM, Mar 10, 2026

4. **Queries / Discussion**

Please post on the Aula Global Forum for everyone's benefit.

5. **Solved Practices Update:** Will not be available, refer to tips and guidance on github.

Tips from Practice Submission 1



I am working on grading your first practice notebook. I will get your submissions back to you as soon as possible. Any announcements or updates will be posted on Aula Global.

1. Meaningful variable names
2. Use code comments only to explain your code
3. Use markdown cells to explain written answers
4. Make sure written answers completely answer the question

Outline



In **Module I**, we will focus on **Regression**

1. “Fairness via Unawareness”
2. Omitted Variable Bias
3. Fairness Criteria for Regression
 - a. Counterfactual Fairness
 - b. Statistical (or Demographic) Parity

Datasets

1. Discrimination in Salaries
2. Law School Success

Discrimination in Salaries Data

Sex	Rank	degree	Years of Experience	Salary
male	full	doctorate	35	36350
male	associate	doctorate	19	24750
male	assistant	doctorate	23	19175
female	full	doctorate	27	26775
female	full	masters	32	24900
...	...	...	...	...

From a small midwestern college from United States in the early 1980s.

N = 52 Faculty

- ▶ 38 Male (73%)
- ▶ 14 Female (27%)

Underlying Biases

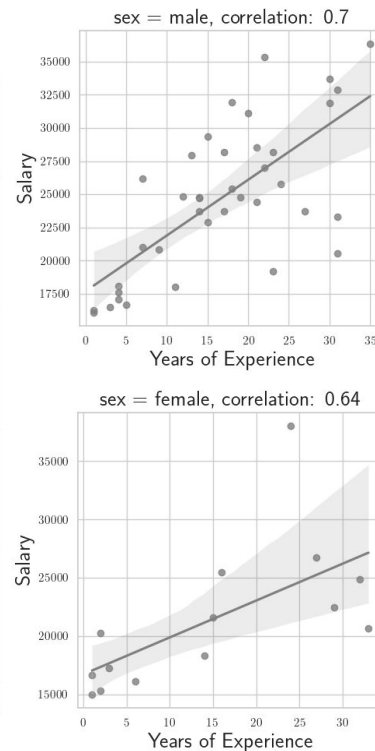
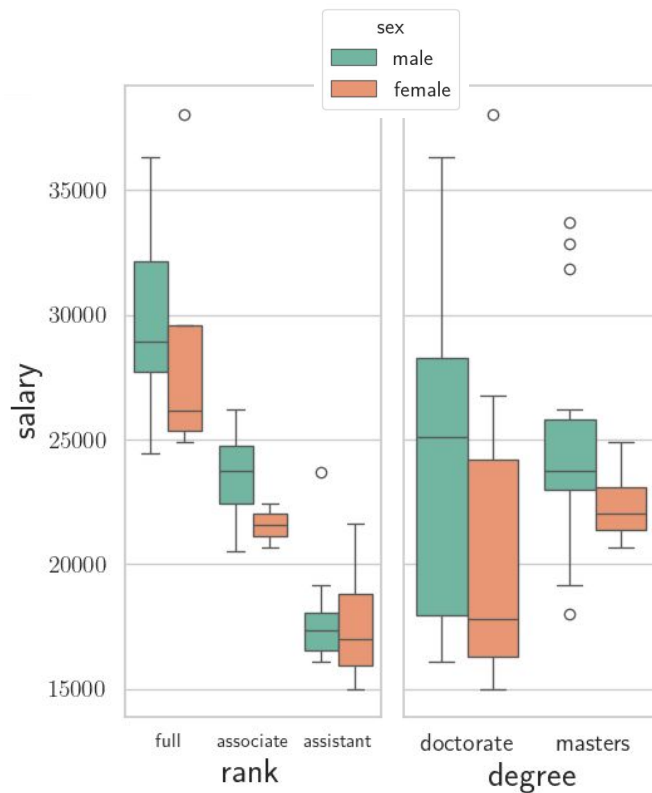
Faculty with Sex = Male

- ▶ 26% assistant, 32% associate, 42% full
- ▶ 37% masters, 63% doctorate

Faculty with Sex = Female

- ▶ 57% assistant, 14% associate, 29% full
- ▶ 29% masters, 71% doctorate

What can we observe?



A Test of Fairness via "Unawareness"

Linear Regression

$$\text{Salary } y = w^T x = w_0 + w_1 * \text{Rank} + w_2 * \text{Degree} + w_3 * \text{Experience} + w_4 * \text{Sex}$$

assistant = 1
associate = 2
full = 3

masters = 1
doctorate = 2

male = 0
female = 1

Should we remove sex from the model?

Should we build separate models?

Omitted Variable Bias

True Model ▶ $y = a_0 + a_1x_1 + a_2x_2 + \beta s$

Fitted Model ▶ $y' = b_0 + b_1x_1 + b_2x_2$

where **s** : sensitive attribute and $\beta \neq 0$

b_0, b_1, b_2 will be **biased** due to the **omission** of **s**

when **s** is **correlated** with x_1 or x_2 or both...

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$$\text{Salary } y = w^T x = w_0 + w_1 * \text{Rank} + w_2 * \text{Degree} + w_3 * \text{Experience} + w_4 * \text{Sex}$$

assistant = 1
associate = 2
full = 3

masters = 1
doctorate = 2

male = 0
female = 1

- | | | |
|-------------|---|--------------------------------------|
| Aware Model | ▶ | Trained on all Attributes |
| Blind Model | ▶ | Trained on all Attributes except Sex |
| M Model | ▶ | Trained on Data where Sex = Male |
| F Model | ▶ | Trained on Data where Sex = Female |